Tony (Haoze) Guo

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Education

University of Wisconsin - Madison Madison, WI

B.Sc. in Computer Engineering double major in Computer Science *Sep 2023 – May 2027*

- GPA: 3.78/4.00; Major GPA: 3.84/4.00; Honors in Computer Science
- Relevant Courseworks: Human-Computer Interaction, Computer Vision, Artificial Intelligence, Data Structure and Analysis, Computer Architecture, Software Engineering, Algorithm, Database Management

Publications & Manuscripts

2027 **Haoze Guo**, et al.
Parallax: Addressing Grounding Asymmetry in Human–Agent Collaboration Across Conversation and Applications
In submission to **CHI 2027**.

2026 **Haoze Guo**.
ConsentDiff at Scale: Longitudinal Audits of Web Privacy Policy Changes and UI Frictions [Link]
In Proceedings of Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems **(CHI EA 2026)**.

Haoze Guo, and Ziqi Wei.
Hidden-in-Plain-Text: A Benchmark for Social-Web Indirect Prompt Injection in RAG [Link]
In Proceedings of the ACM **Web Conference 2026** (WWW 2026).

Mohammed Almutairi, Charles Chiang, **Haoze Guo** et al., and Diego Gómez-Zará.
Simulating Teams with LLM Agents: Interactive 2D Environments for Studying Human-AI Dynamics [Link]
In submission.

2025 Mohammed Almutairi, Charles Chiang, **Haoze Guo** et al., and Diego Gómez-Zará.
VIRT-LAB: An AI-Powered System for Flexible, Customizable, and Large-scale Team Simulations [Link]
Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology **(UIST 2025)**.

Mohammed Almutairi, Charles Chiang, **Haoze Guo** et al., and Diego Gómez-Zará.
Enabling User-Created Multi-Agent Simulations: Interactive and Customizable 2D Environments to Study Team Dynamics with LLM Agents [Link]
In proceeding of the 39th Conference on Neural Information Processing Systems (**NeurIPS 2025**) Workshop: Scaling Environments for Agents (SEA).

Research Experience

Yifang's Lab Jan. 2026 – Present

Undergraduate Research Assistant ***PI: Dr. Yifang Wang*** *Remote*

- Designed and developed an interactive human–AI system for maintaining task context and grounding across shared-workspace workflows, with prototype applications in literature review and network visualization.
- Conducted formative user research and translated findings on context loss, workflow continuity, provenance, and user control into system design requirements and grounding mechanisms.
- Implemented and iterated on system interfaces and backend infrastructure for semantic read-back, task-state management, and selective incorporation of application-side evidence into AI interactions.

People and Robots Laboratory Oct. 2025 – Present

Undergraduate Research Assistant ***PI: Prof. Bilge Mutlu*** *Madison, WI*

- Developing an LLM version-control system for managing, comparing, and reproducing model behavior across model and prompt versions.

- Designed interface components and backend infrastructure to support systematic evaluation and inspection of behavioral changes across LLM configurations.

Research In Networks and Groups (RINGZ) Lab May. 2025 – Aug. 2025
Undergraduate Research Assistant ***PI: Dr.Diego Gómez-Zará*** *Notre Dame, IN*

- Led optimization and backend development of VirT-Lab, a **multi-agent** simulation platform powered by **LLMs** for modeling dynamic **team** behavior in spatial environments.
- Implemented scalable event scheduling and LLM-based reasoning pipelines using **Python**, **WebSockets**, and **GPT-4o-mini**, enabling real-time coordination across autonomous agents.
- Enhanced system performance by reducing LLM response latency by **35%** and memory usage by **20%**

Professional Experience

Oshkosh Corporation × University of Wisconsin–Madison Sep. 2026 – Present
AI & Data Engineering *Madison, WI*

- Designing a **Databricks-based AI-ready data pipeline** for heterogeneous manufacturing and supply-chain data, including PDFs, spreadsheets, presentations, images, and structured system exports.
- Building a **semantic data layer** with Python, Apache Spark, SQL, and Delta Lake to extract document structure, metadata, revisions, and source lineage for downstream AI-agent retrieval.
- Developing **embedding/vector-search and RAG pipelines** with evaluation metrics for data completeness, extraction quality, traceability, retrieval quality, and AI response reliability.

ByteDance Apr. 2023 – Sep. 2023
Software Development Engineering *Beijing, China*

- Developed and optimized a scalable code-hosting platform using C/C++ and Java, improving platform reliability by **50%** and performance by **40%**.
- Implemented automated monitoring and IaC best practices, reducing system downtime by **30%**, incident response time by **50%**, and deployment time by **60%**.

University of Wisconsin - Madison Feb. 2025 - May 2025
Data Analyst *Madison, WI*

- Conduct comprehensive data cleaning, and exploratory data analysis (EDA) using Python and R libraries such as **pandas**, **NumPy**, **dplyr**, and **ggplot2**, ensuring data accuracy and integrity for subsequent analysis.
- Develop **SQL** queries to efficiently extract and aggregate relevant datasets from relational databases, optimizing query performance to handle large-scale data retrieval tasks.

Projects

Parallax (advised by Dr. Yifang Wang) | *Human-AI Collaboration, LLM Agents* Jan. 2026 – Sep. 2026

- Designed a **persistent TaskFrame architecture** for tracking user intent, workflow stage, constraints, required evidence, and agent state across multi-application workflows.
- Built a **multi-layer semantic read-back pipeline** that maps raw user interactions into semantic events and task-conditioned evidence with artifact-level provenance.
- Implemented **evidence-gap routing and selective grounding** to retrieve relevant prior interactions, artifacts, and application state for LLM-based task continuation.

VirT-Lab (advised by Dr. Diego Gómez-Zará) | *JavaScript, Multi-Agent Systems* Jun. 2025 – Aug. 2025

- Led optimization and backend development of VirT-Lab, a **multi-agent** simulation platform powered by **LLMs** for modeling dynamic **team** behavior in spatial environments.
- Implemented scalable event scheduling and LLM-based reasoning pipelines using **Python**, **WebSockets**, and **GPT-4o-mini**, enabling real-time coordination across autonomous agents.
- Enhanced system performance by reducing LLM response latency by **35%** and memory usage by **20%**

Academic Services

External Reviewer: IUI Paper (2027) & CHI Poster (2026) & IUI Papers & Poster(2026), ICWSM Paper(2026)
Program Committee: CHI, IUI, ICWSM

Technical Skills

Languages: Python, JavaScript, Java, MATLAB, SQL, C/C++, R, C#, TypeScript, Go
Developer Tools: Git, Linux, Google Cloud Platform, AWS(Lambda), PyCharm, Unity Hub
Frameworks & Libraries: React, Express, Node.js, Figma, Vue.js, PyTorch